



মণিপুর ইনস্টিটিউট অফ টেকনোলজী
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(A Constituent College of Manipur University)

A PROJECT
ON
MEDICAL IMAGE ENHANCEMENT BASED
ON DENOISING

A THESIS SUBMITTED AS A PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF THE DEGREE OF
MASTER OF TECHNOLOGY [M.TECH] IN COMPUTER SCIENCE
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ACRONYMS

MSE	Mean Square Error
PSNR	Peak Signal to Noise Ratio
SNR	Signal to Noise Ratio

ABSTRACT

Image enhancement is to process an image so that result is better than original image for specific application. Medical image processing is a hard task in hand. Most of the medical images have very low contrast or full of noise and the challenge is to sharpen them and make them easier in processing. Based on Denoising i.e. noise addition (The Gaussian noise is added to the original image) and then noise removal (the collected images are subjected to Noise Removal using the Wiener filtering method). Also the edge based image enhancement is done using the sharpening method. The features are collected from the image based on the patterns. The resulted image was better when it compared to the original image. The goodness of image are determined by measuring the quality of both original and resulted image using quality assessment parameters which include Mean Square Error(MSE), Peak Signal to Noise Ratio(PSNR) and Signal to Noise Ratio(PSNR).

CERTIFICATE

This is to certify that the thesis entitled **Medical Image Enhancement based on Denoising** submitted by **Dingku Singh Oinam** to the Department of Computer Science and Engineering, Manipur Institute of Technology, Manipur University towards partial fulfillment of the requirements for the award of the degree of Master of Technology [M.tech] in Computer Science and Engineering is a bonafide research work carried out by him under our supervision and guidance. This work has not been submitted previously for any other degree of this or any other University. It is further certified that the candidate has completed with all the formalities as per requirements of Assam University. We recommend that the thesis may be placed before the examiner for consideration of award of the degree of this University.

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Date.....

Place.....

CERTIFICATE

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(Dr. N. Basanta Singh)

Principal,

Manipur Institute of Technology, Imphal

Date.....

Place.....

CHAPTER ONE

INTRODUCTION

1.1 Preface

Image enhancement techniques are used to improve image quality or extract the fine details in the degraded images. It is to modify attributes of an image to make it more suitable for given task.

1.2 Problem Statement

Medical images are one of the fundamental images, because they are used in more sensitive field which is a medical field. They are ought to be understand because it may contain noise, it may be dark or so white or it may become blurred, so image enhancement will applied to enhance medical image.

1.3 Research Importance

Diagnosis depends only on radiograph which are captured by specialist. From this point we need to suppose scenarios which make image easier for visual interpretation to the doctors whom are not forced to do radiography more than one time which ensures timeserving to the doctors and patients, speed and accuracy in making decision.

1.4 Literature Review

1.4.1 Image Enhancement

First it takes input .png or .jpg image. The image are converted into grayscale image. Create Random Initial State: add noise in image by change their contrast and bit map value. Evaluation fitness is done through noisy section of image. This proposed method gives better result than all the other individual method. It reflexes an effort to understand the better quality of image by increasing the contrast of an image. It is observed that genetic algorithm can play role efficiently to analyze image.

1.5 Research Objectives

The main objectives of this research are to:

- Increase research in medical image area.
- Increase the ability of enhancing medical image for patients.
- Increase the enhancing time of processing because it is considered a very sensitive area (medical area).
- Improve the visual quality of an image. Contrast increment, elimination of noise and enlightenment of details.

1.6 Research Methodology

1.6.1 Genetic Algorithm

Genetic Algorithms encode a potential solution to specific problem on a simple chromosome like data structure and apply recombination operators to these structures so as to preserve critical information.

1.6.2 Matlab Software

Matlab is considered as solution of image processing problem. The power that MATLAB brings to digital image processing an extensive set of functions for processing multidimensional arrays. The image processing toolbox is a collection of functions that extend the capability of the MATLAB numeric computing environment.

1.7 Research Framework

The steps of program can be illustrated in points below for enhancing the medical image using genetic algorithm:

- 1) Uploaded original image.
- 2) Convert original to gray image.
- 3) Determining the image edges.
- 4) Add noise to gray image in order to create random population.
- 5) Gray image and noisy image is crossed over (Recombined).

CHAPTER TWO

DIGITAL IMAGE PROCESSING

2.1 Image Processing Task

Image processing and pattern recognition has been studied extensively in computer science. Image processing task may be accomplished in steps below:

- **Acquiring the image:** Digital image produced from a paper envelope. This can be done using either a camera, or a scanner.
- **Pre-processing:** This is step taken before the major image processing task. The problem here is to perform some basic tasks in order to render the resulting image more suitable for the job to follow. In this case it may involve enhancing the contrast, removing noise, or identifying regions likely to contain the postcode.
- **Segmentation:** Here the postcode is extracted; in other words it is the process which done by extract from the image that part of it which contains just the postcode.
- **Representation and Description:** These terms refer to extracting the particular features which allowed differentiating between objects. It is done by looking for curves, holes and corners which allowed distinguishing the different digits which constitute a postcode.
- **Recognition and interpretation:** This means assigning labels to objects based on their descriptors (from the previous step), and assigning meanings to those labels. So identifying particular digits, and interpreting a string of four digits at the end of the address as the postcode.

2.1.1 Digital Image Representation

Digital image processing refers to process digital image by means of a computer. Image processing takes image as input, process it and produces the output. The image can be defined as an array, or a matrix, square pixel arranged in rows and columns.

2.1.2 Digitizing Image(Sampling)

An image may be continuous with respect to the x- and y-coordinates, and also in amplitude converting such an image to digital form requires that the coordinates, as well as the amplitude, be digitized. Digitizing the coordinate values is called sampling; digitizing the

amplitude values is called quantization. Thus, when x , y , and the amplitude values of f are all finite, discrete quantities, we call the image a digital image.

2.1.3 Image Types

Image types include:

- **Binary Images:**
Each pixel is just black or white. Since there are only two possible values for each pixel, the needs are only for one bit per pixel. Such images can therefore be very efficient in terms of storage.
- **Gray-scale Images:**
Gray-scale image is a data matrix whose values represent shades of gray. When the elements of a gray-scale image are of class `uint8` or `uint16`, they have integer values in the range $[0, 255]$ or $[0, 65535]$, respectively. If the image is of class `double` or `single`, the values are floating-point numbers. Values of `double` and `single` gray-scale images normally are scaled in the range $[0, 1]$, although other ranges can be used.
- **Indexed Images:**
The image has an associated colour map, or colour palette, which is simply a list of all the colors used in that image. Each pixel has a value which does not give its colour, but an index to the colour in the map. It is convenient if an image has 25 colors or less, the index values will only require one byte to store. Some image file formats allow only colors or fewer in each image.
- **Truecolor Images:**
Truecolor image also named RGB images (Red, Green, and Blue). It is stored as an m -by- n -by-3 data array that defines red, green, and blue color components for each individual pixel. Truecolor images do not use a palette. The color of each pixel is determined by the combination of the red, green, and blue intensities stored in each color plane at the pixel's location. Graphics file formats store RGB images as 24-bit images, where the red, green, and blue components are 8 bits each. This yields a potential of 16 million colors. The precision with which a real-life image can be replicated has led to the nickname "truecolor image".

2.2 Medical Image Processing

The efforts on medical image enhancement have been focused mostly to improve visual perception of images that are unclear because of blur. Edges are the representations of the discontinuities of image intensity functions. For processing these discontinuities in an image a good edge enhancement technique is essential reduces complexity and makes the images look sharper than they really are.

2.3 The Quality Of The Medical Image

The quality of the medical image is as good as the clarity of the specific information sought for in the image by the observing physician. However, there is a need to define objective image quality features in order to assess the quality of the image produced by an imaging system or by processing techniques. The quality of the medical image depends on and assessed by three parameters: sharpness, contrast and noise. The image quality assessment aims to use computational models so as to measure the image quality consistently with subjective evaluations.

2.4 Image Enhancement

Image enhancement technique is use to convert the original image into the better image. The input image can be from any image capturing device. There are various methods which can enhance the original image without losing its original good properties. Digital image enhancement techniques provide a multitude of choices for improving the visual quality of images. Image enhancement refers to processing an image so that the result is more suitable for a particular application. Examples include: sharpening or de-blurring an out of focus image, highlighting edges, improving image contrast, or brightening an image and removing noise. In image enhancement process one or more attributes of image are modified. The enhancement methods can broadly be divided into the following two categories:

2.4.1 Spatial Domain Methods

The term spatial domain refers to the image plane itself, i.e. aggregate of pixels of composing an image, and approaches in this category are based on direct manipulation of pixels in an image. Spatial domain methods directly operate on pixels .These techniques are based on gray level mappings, and the type of mapping used for these technique are depends on the criterion chosen for enhancement.

2.4.2 Frequency Domain Methods

In frequency domain methods, the image is first transferred into frequency domain. For example the Fourier transform of the image is computed. All the enhancement operations are performed on the Fourier transform of the image and then the Inverse Fourier transform is performed to get the resultant image. These enhancement operations are performed in order to modify the image brightness, contrast or the distribution of the grey levels. As a consequence the pixel value (intensities) of the output image will be modified according to the transformation function applied on the input values.

2.5 Image Noise

Noise is an important factor that influences image quality which is mainly produced in the processes of image acquirement and transmission. Noise reduction is necessary for us to do image processing and image interpretation so as to acquire useful information that we want. Because the working status of image transmitter is influenced by varied of factors, such as the environment of image acquired, different noises can be dealt with in different ways. Through the analysis of image spectrum, its difference can help us to choose different methods to do noise reduction while the information of the image is reduced to be the most.

Image noise is generally regarded as an undesirable by-product of image capture. Although these unwanted fluctuations became known as "noise" by analogy with unwanted sound they are inaudible and such as dithering. There are many types of noise which includes:

- 1) Gaussian noise.
- 2) Salt-and-pepper noise.
- 3) Poisson noise.
- 4) Speckle noise.

2.6 Image Restoration

This may be considered as reversing the damage done to an image by a known cause, it may include: removing of blur caused by linear motion, removal of optical distortion and removing periodic interference. The distortion in image can be modeled as noise or blur or a degradation function. Unfortunately all images are more or less blurry. This is due to the reason that there is a lot of interference in the camera as well as in the environment. Blurring of an image can be caused by many factors

such as movement during the capture process, using wide angle lens, using long exposure times, etc.

The field of image restoration (sometimes referred to as image deblurring or image deconvolution) is concerned with the reconstruction or estimation of the uncorrupted image from a blurred and noisy one. Essentially, it tries to perform an operation on the image that is the inverse of the imperfections in the image formation system. Some image restoration techniques are:

1. Median filter:

The median filter is used to remove the salt and pepper noise. It has capability with considerably less blurring than linear smoothing filters of the similar size. In other words median filtering is a widely used and very important technique of filtering and best known for its excellent noise reduction ability from the images. By the filtering it keeps the edges while removing the noise. This makes the image not to blur as other smoothing methods.



Fig 2.1 De-noise image using median filter.

2. Adaptive filter:

Adaptive filter is type of linear filter that has transfer function controlled by variable parameter. Adaptive filter use the color and gray space for removal of impulsive noise in images. All processing is done on the basis of color and gray space. Adaptive filter are used to remove the effect of speckle noise. This can provide the best noise suppression results and better preserve edges, thin lines and image details and yield better image quality in comparison to other filters.

3. Linear filter:

In linear filter each pixel is replace with linear combination of its neighbors. Image processing operations are implemented with linear filter include sharpening, smoothing and edge enhancement. In linear filter output change linear with input.

With the help of linear filter we can easily remove the noise from the image. This filter can be implemented on salt and pepper noise and Gaussian noise.

2.7 Enhancement Performance Parameters

It includes the parameters which used to estimate the degree of enhancement of resulted image when we compare it to the original image.

2.7.1 Mean Square Error (MSE)

The MSE is the cumulative square error between the encoded and the original image defined by:

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} ||f(i,j) - g(i,j)||^2 \dots\dots\dots (2.1)$$

Where:

f is original image.

g is uncompressed image.

m, n represent dimension of image.

i, j represent the index of the row and column

Thus, MSE should be as low as possible for effective compression.

2.7.2 Peak Signal to Noise Ratio (PSNR)

PSNR is the ratio between maximum possible power of a signal and the power of distorting noise which affects the quality of its representation. It is defined by:

$$PSNR = 20 \log_{10} \frac{MAXf}{\sqrt{MSE}} \dots\dots\dots (2.2)$$

Where:

MAXf is a maximum signal value that exists in our original image.

“known to be good” image.

$\sqrt{MSE}$ is root mean square error.

CHAPTER THREE

THE PROPOSED METHOD

In this paper, the medical image dataset are implemented as input. The input images are captured in ".jpg" or ".png" format. A grayscale image is created from the photographs. The image is then given a layer of Gaussian noise. Hence the level of the noise can be given by the user. The collected image are subjected to Noise Removal using the Wiener filtering method. The edge based image enhancement is done using the sharpening method.

And to check the quality of the image, MSE and PSNR will be estimate. Image enhancement is to process an image so that result is better than original image for specific application. The difficulty is to brighten them and make them simpler to process because the majority of medical photos have extremely poor contrast or are jam-packed with noise.. Based on Denoising i.e. noise addition (The Gaussian noise is added to the original image) and then noise removal (the collected images are subjected to Noise Removal using the Wiener filtering method). Also the edge based image enhancement is done using the sharpening method. The features are collected from the image based on the patterns. When compared to the original image, the outcome was better.

3.1 Overview Of Image Denoising

Image denoising, whose main purpose is to estimate the original image by suppressing noise from a noise-contaminated version of the image, is one of the fundamental problems in the field of image processing and computer vision. Different intrinsic (sensor-related) and extrinsic (environment-related) circumstances, which are frequently impossible to prevent in real-world applications, can result in image noise. Due to the need to recover the original image content for effective performance, image denoising is used in a variety of applications, including image restoration, visual tracking, image registration, image segmentation, and image classification. Although various algorithms have been put forth for the goal of image denoising, the issue of image noise suppression still remains unsolved, particularly when the photos are taken in unfavourable conditions with a high degree of noise.

3.2 De-noising

Image denoising is a classic problem and has been studied for a long time. However, it remains a challenging and open task. The main reason for this is that from a mathematical perspective, image denoising is an

inverse problem and its solution is not unique. In recent decades, great achievements have been made in the area of image denoising. Few steps to enhance the quality of Image based on denoising are given below:

1. Noise Addition: In this step, the Gaussian noise is added to the image. Hence the level of noise can be given by the user.
2. Noise Removal: The collected images are subjected to Noise Removal using Wiener filtering method.
3. Image Super Resolution: In this process the Upscaling will be done. Hence the number of row and column in the image is increased to improve the resolution of the image. Also the edge based image enhancement is done by using the sharpening method.

3.3 Gaussian Noise

Gaussian noise, named after Carl Friedrich Gauss, is a term from signal processing theory denoting a kind of signal noise that has a probability density function (pdf) equal to that of the normal distribution (which is also known as the Gaussian distribution). In other words, the values that the noise can take are Gaussian-distributed. The probability density function of a Gaussian random variable is given by:

$$p_G(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

where z represents the grey level, μ the mean grey value and σ its standard deviation.

Algorithm 1: Algorithm for adding Gaussian Noise	
1:	<code>r=imread('abc.png');</code> %input image
2:	<code>axes(handles.axes1);</code>
3:	<code>imshow(r);</code>
4:	<code>n=imnoise(r,'Gaussian',0.02),</code> %add noicelevel 0.02
5:	<code>axes(handles.axes2);</code>
6:	<code>imshow(n)</code>

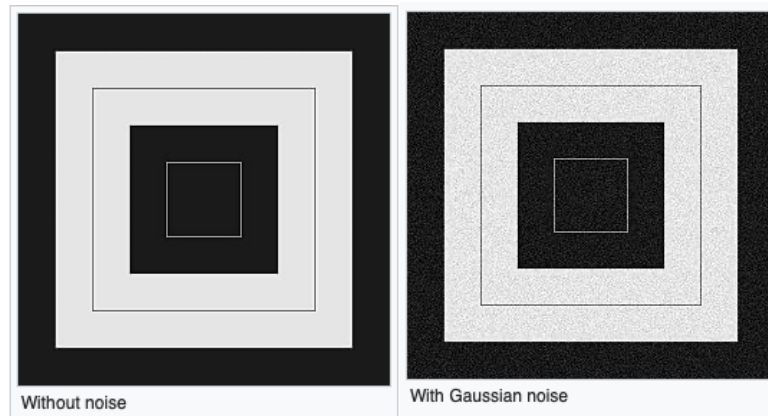


Fig 3.1 Image without Gaussian Noise and Image with Gaussian Noise

3.4 Wiener Filter

By applying linear time-invariant (LTI) filtering to an observed noisy process, the Wiener filter can be used to estimate a desired or target random process. Additive noise and established stationary signal and noise spectra are assumed. The mean square error between the intended process and the estimated random process is reduced by the Wiener filter.

By employing a related signal as an input and filtering that known signal to obtain the estimate as an output, the Wiener filter aims to compute a statistical estimate of an unknown signal. For instance, the known signal might be made up of a potentially valuable unknown signal that has been tampered with by additive noise. By removing the noise from the distorted signal, the Wiener filter can estimate the underlying signal of interest. The Wiener filter is based on a statistical methodology, and the article on the minimal mean square error (MMSE) estimator provides a more statistical explanation of the theory.

Algorithm2: Algorithm for Wiener Filter	
1:	<code>x=imread('bcd.png');</code> %input Gaussian noise image
2:	<code>imshow(x);</code>
3:	<code>b=wiener2 (x,[5 5]);</code> %Wiener filter
4:	<code>imshowpair(x,b,'montage')</code>

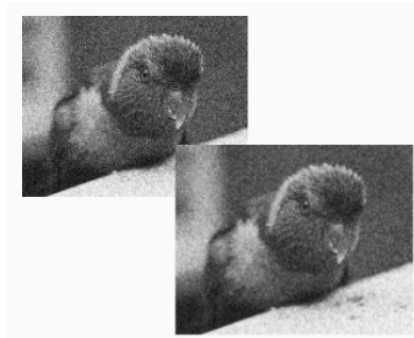


Fig 3.2 Gaussian Noise Image and Wiener Filtered Image

3.5 Image Enhancement using Sharpening method

Image sharpening is an effective tool for highlighting texture and grabbing the attention of the viewer. Whether you are aware of it being used or not, it is also a requirement for all digital photos at some time. For instance, the blur that results from digital camera sensors and lenses must be corrected. However, not every sharpening method is the same. Unattractive sharpening artefacts may occur if the process is carried out too vigorously. However, when done properly, sharpening can frequently enhance perceived image quality much more than purchasing a high-end camera lens. Given that human perception is extremely sensitive to an image's edges and fine details, which are predominantly made up of high frequency components, attenuating or completely eliminating the high frequencies can have a significant negative impact on an image's visual quality. On the other hand, boosting an image's high-frequency elements results in a visual quality improvement. Any enhancing method that brings out an image's finer features and edges is referred to as sharpening. In order to improve the local contrast and sharpen the images, image sharpening is frequently employed in the printing and photographic industries.

Algorithm2: Algorithm for Sharpening Image	
1:	<code>x=imread('cde.png');</code> % input Wiener Filter image
2:	<code>figure;</code>
3:	<code>imshow(x);</code>
4:	<code>sharp_image =imsharpen(image,'Amount',0.8);</code>
5:	<code>sharp_image=adapthisteq(sharp_image);</code>
6:	<code>imshow(sharp_image);</code>
8:	<code>title('Result Image');</code>

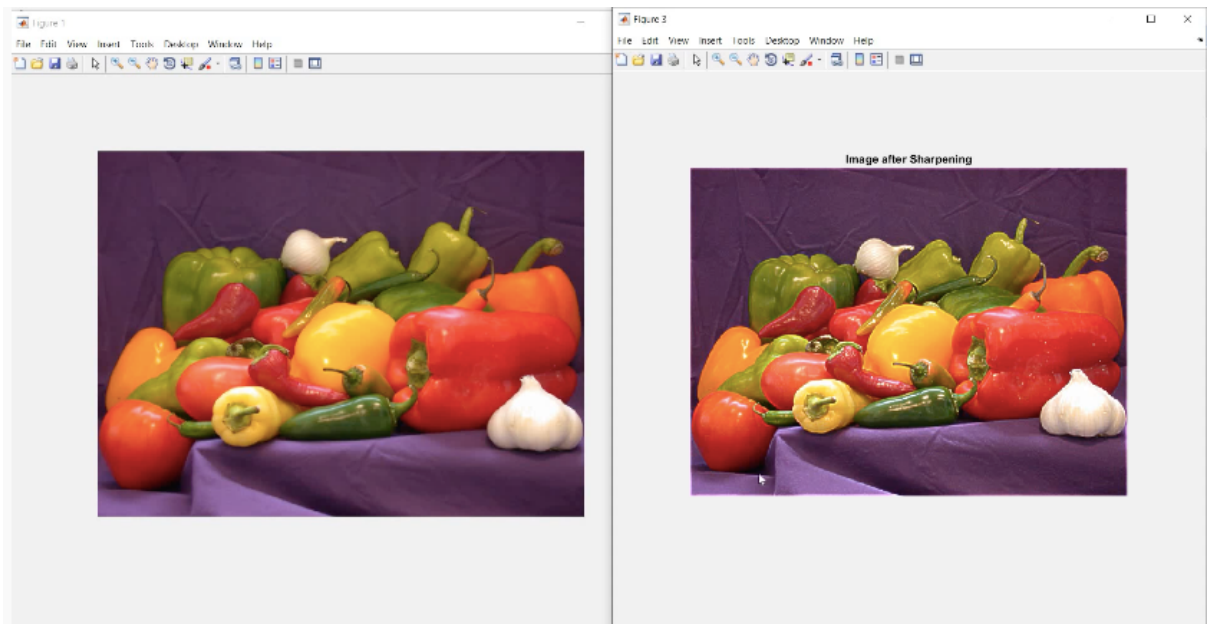


Fig 3.3 Wiener Filter Image, and Enhance Image

3.6 Image Denoising Applications

Image denoising is a key component in a variety of applications, including image restoration, visual tracking, image registration, image segmentation, and image classification, where recovering the original image content is essential for effective performance.



Fig 3.4 Original Image, Image with Gaussian Noise, and Image After Wiener Filter

CHAPTER FOUR

RESULTS

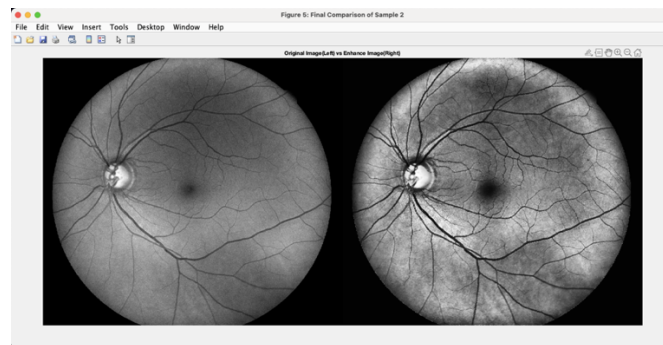
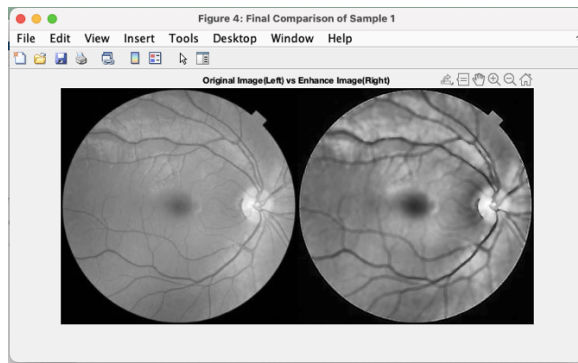


Fig 4.1 Final Comparison between the original and enhance image.

4.1 Digital Image Processing

Image processing involves changing the nature of an image in order to:

- 1) Improve its pictorial information for human interpretation.
- 2) Render it more suitable for autonomous machine perception.

4.2 Description Of Data

Data are taken by specialist from various resources which include (Internet)for research purpose.

They are two samples of image which includes: Glaucoma(A), Glaucoma(B)

TABLE 4.1 Description of Data

Sample number	Original image	Dimension
1.	S1.jpeg	<i>500x500</i>
2.	S2.jpeg	<i>1644x840</i>
3.	S3.jpeg	<i>1800x2700</i>

4.3 Steps Of Processing

The processing steps for enhancing image are:

- The Medical Image dataset are implemented as input. Image uploaded from the path which you select.
- The images are converted into grayscale image.
- Gaussian Noise is added to the original image. Then give us an output of the image.
- The Noise is then remove using Wiener filter method.
- We determine the edges of image, in this process the Upscaling is done.
- Finally we get the enhance image.
- Then to check the quality of the image, PSNR and MSE is estimate.

4.4 Case Study:

Glaucoma Image:

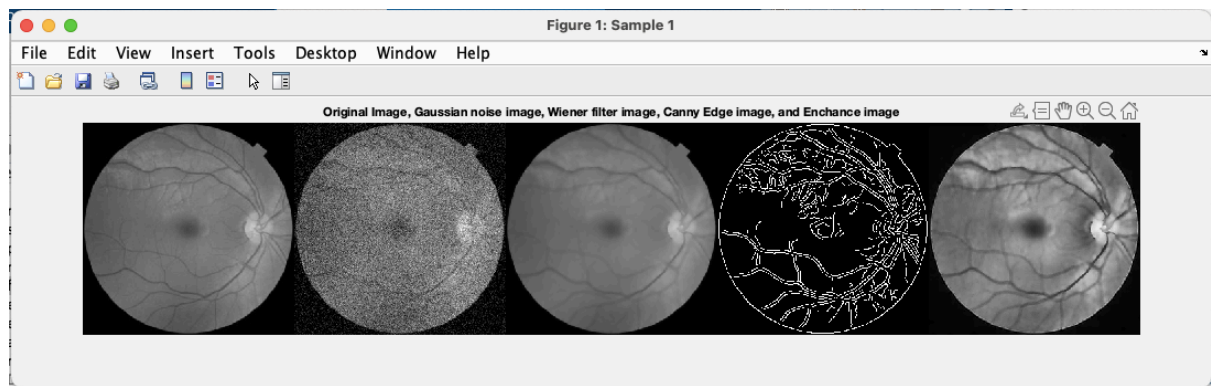


Fig 4.2 Original Image, Gaussian Noise Image, Wiener Filter Image, Edge Image, and Enhance Image of Sample 1.

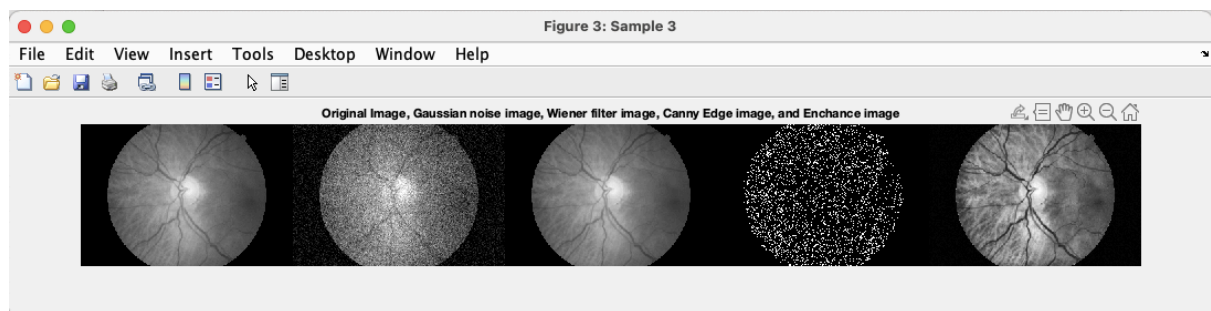


Fig 4.3 Original Image, Gaussian Noise Image, Wiener Filter Image, Edge Image, and Enhance Image of Sample 2.

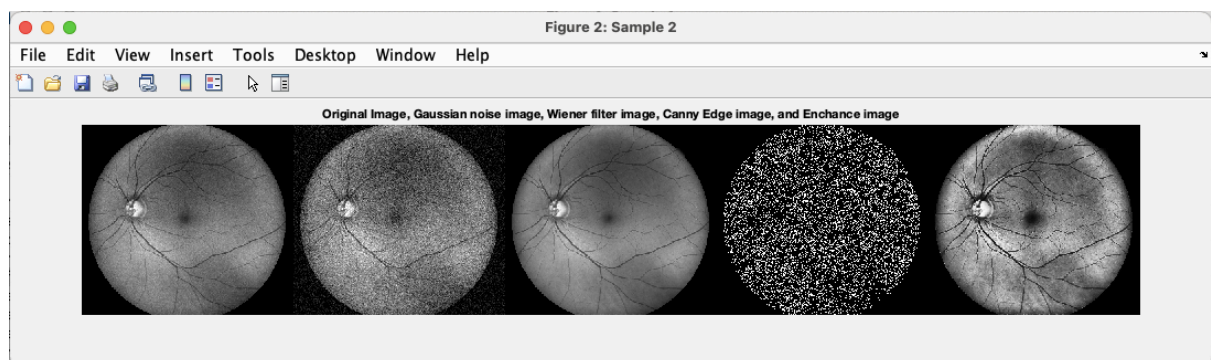


Fig 4.4 Original Image, Gaussian Noise Image, Wiener Filter Image, Edge Image, and Enhance Image of Sample 3.

4.5 Density of Noise

It is the number of white and black pixels which are added to make image noisy. The algorithm was used a default value (0.01).

TABLE 4.2 Density of Noise

Sample	Noise Density
1.	0.01
2.	0.01
3.	0.01

4.6 Quality of the Image

TABLE 4.3 Quality of the Image

Sample	MSE		PSNR		SNR	
	Original	Enhance	Original	Enhance	Original	Enhance
1.	589.5697	356.2702	20.4255	22.6130	12.3192	14.5067
2.	587.1295	592.1944	20.4435	20.4062	12.4000	12.3627
3.	554.1201	295.9482	20.6948	23.4186	12.1745	14.8984

Hence, the enhance image is better than the original image.

CHAPTER FIVE

CONCLUSION AND FUTURE WORK

5.1 Conclusion

So, we conclude that we obtain a better quality image as compared to the original image. To confirm the result is compared to some standards. The experiment result is satisfactory. Besides, experiment results demonstrated that the enhanced images are suitable for different application field. And we can further do more research in this area.

Some experimental results after enhancement based on denoising, where we can see clearly how effectively it works.

5.2 Future Work

Here we see just visual comparison. Later, we will represent this comparison in statistical way. Moreover, in future, we were going to combine some other algorithm and to compare those result with this one. We would like to include other method like swarm intelligence or ant colony method to improve the result.

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